Multimedia Signal Processing - 2nd Module

September, 12 2016

1. **Question (6 pts)**
   Consider the following sequence of samples:

$$x = \{10,\ 12,\ 15,\ 16,\ 20,\ 19,\ 18,\ 15,\ 13\}.$$

   (a) Obtain the sequence $\tilde{x}$ encoding x using PCM with the following quantizer:

$$\begin{array}{ll} I_1 = [8, 12), & y_1 = 10 \\ I_2 = [12, 16), & y_2 = 14 \\ I_3 = [16, 20), & y_3 = 18 \\ I_4 = [20, 24), & y_4 = 22 \end{array}$$

   (b) Compute the mean squared error (MSE) introduced by the use of PCM.

   (c) Estimate the pmf of each symbol of $\tilde{x}$ and derive the Huffman coding scheme.

   (d) Obtain the sequence $\hat{x}$ encoding x using a closed-loop DPCM with the following quantizer:

$$\begin{array}{ll} I_1 = [-\infty, -1.5), & y_1 = -3 \\ I_2 = [-1.5, 1.5), & y_2 = 0 \\ I_3 = [1.5, +\infty), & y_3 = 3 \end{array}$$

   Consider that the first symbol is already known by the decoder.

   (e) Compute the mean squared error (MSE) introduced by the use of DPCM.

2. **Question (6 pts)**
   (a) Introduce the concept of scalar quantizer, showing the difference between a midtread and a midrise one.

   (b) Describe the compander/expander operation, highlighting how quantization can benefit from it.

   (c) Introduce the concept of vector quantization.

3. **Question (6 pts)**
   (a) Introduce the concept of perceptual audio coding.

   (b) Describe the MPEG1 psychoacoustic model 1 for global masking threshold computation.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider a video coding scheme.
   - [T] [F] The displaced frame difference is computed for all the frames of a sequence.
   - [T] [F] Motion compensation is the process of predicting motion vectors.
   - [T] [F] It is possible to estimate motion vectors with sub-pixel precision.
   - [T] [F] DCT is used for both inter- and intra-coded frames.

   (b) Consider the JPEG coding standard.
   - [T] [F] DCT is applied separately on red, green and blue channels.
   - [T] [F] Zig-zag scan is used to apply run-length coding to quantized DCT coefficients.
   - [T] [F] 4x4 DCT is used for sub-sampled color components.

   (c) Consider two discrete memoryless sources X and Y.
   - [T] [F] Mutual information $I(X,Y)$ can be negative only for independent sources.
   - [T] [F] Conditional entropy $H(X|Y)$ is always lower than $H(X)$.
   - [T] [F] Joint entropy $H(X;Y)$ can be lower than $H(X)$.

5. **Question (Matlab - 10pts)**
   (a) Load the image `lena512color.tiff` and display it in RGB. From now on, consider the image as only composed by the first two color channels (`HxWx2`).

   (b) Compute the entropies of the first two color channels separately (consider each channel as a monodimensional signal).

   (c) Compute the joint entropy of the first two channels (consider the image as a list of pixels, each one composed by two elements only).

   (d) Implement the following hybrid DPCM - transform coding scheme:
   - The first channel is transform coded using 16x16 DCT and quantizer with $\Delta = 5$.
   - The decoded first channel is subtracted to the second channel to obtain the second channel's residual.
   - This residual is transform coded using 16x16 DCT and quantizer with $\Delta = 5$.
   - Finally the second channel is decoded summing the decoded residual to the decoded first channel.

   (e) Compute for each channel the PSNR achieved after decoding and the coding rate (in terms of entropy).